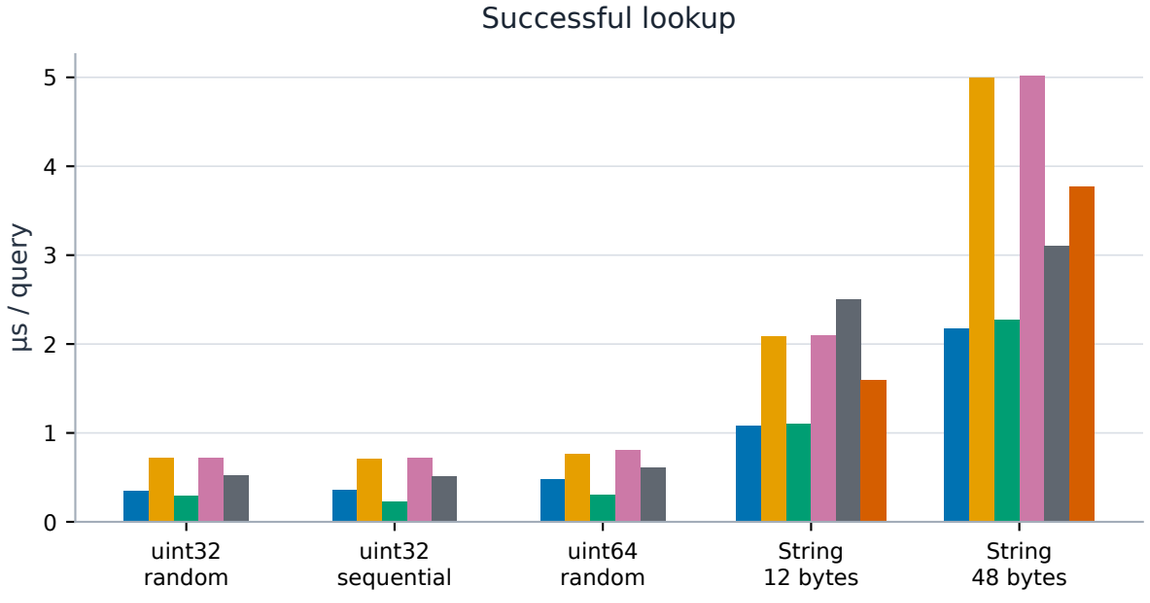
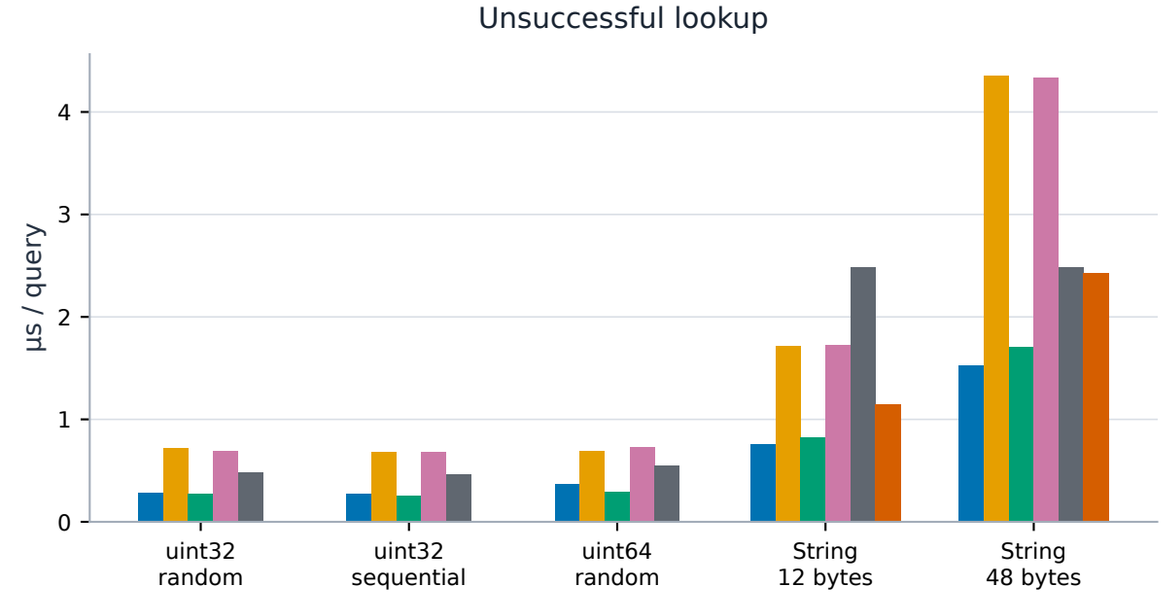
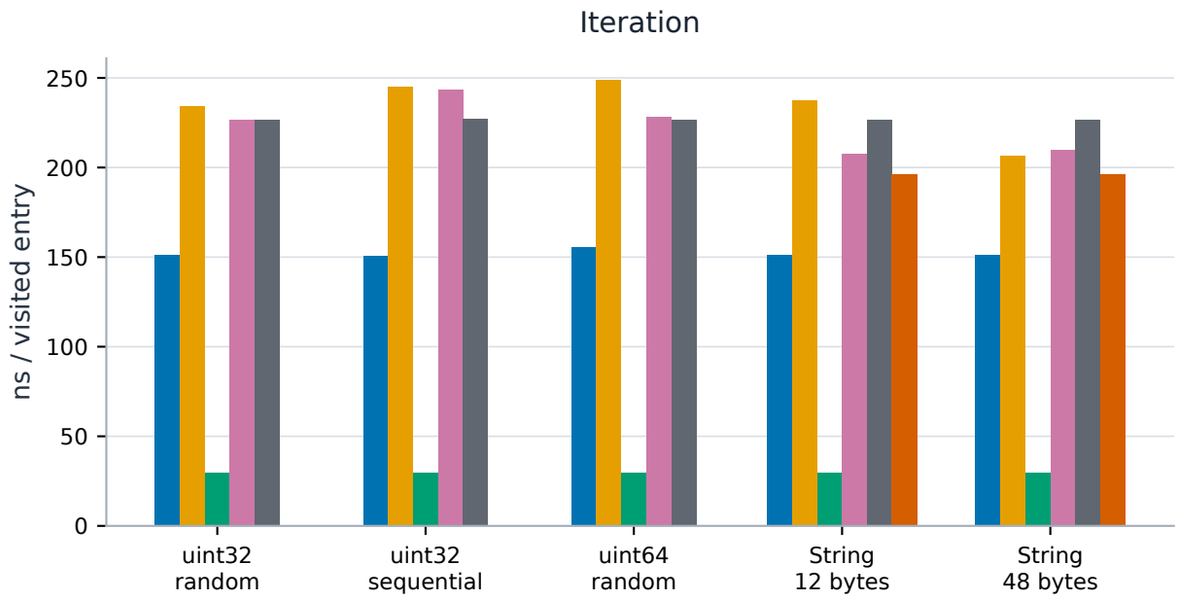
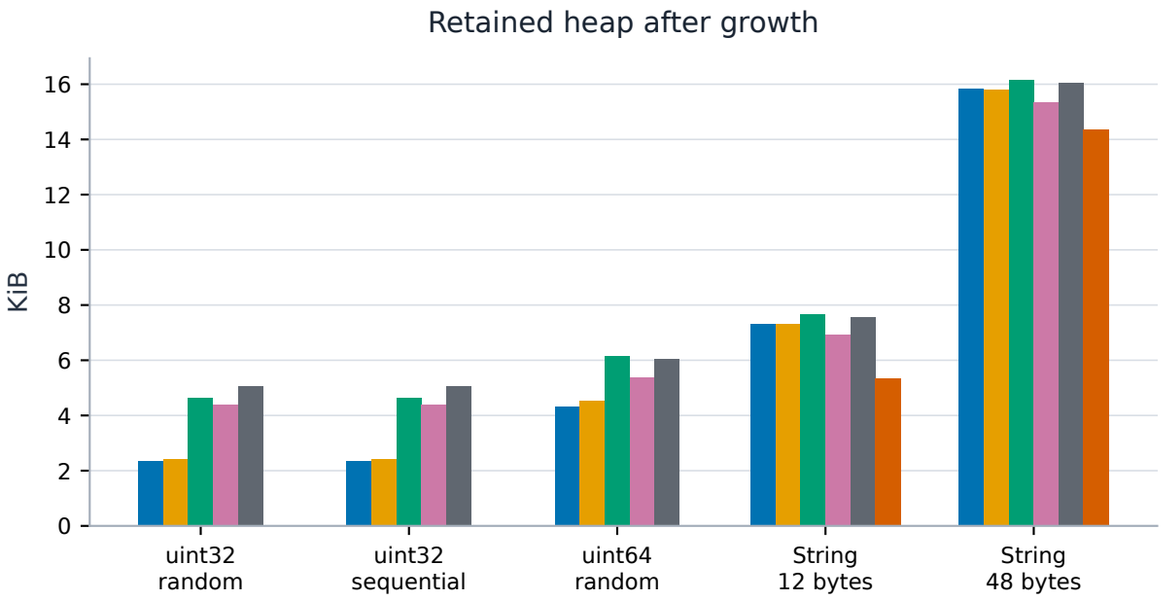
Current map comparison · 128 entries

Default hashers; std::map uses ordering. Heap includes the map object and allocator overhead; it is not peak memory.









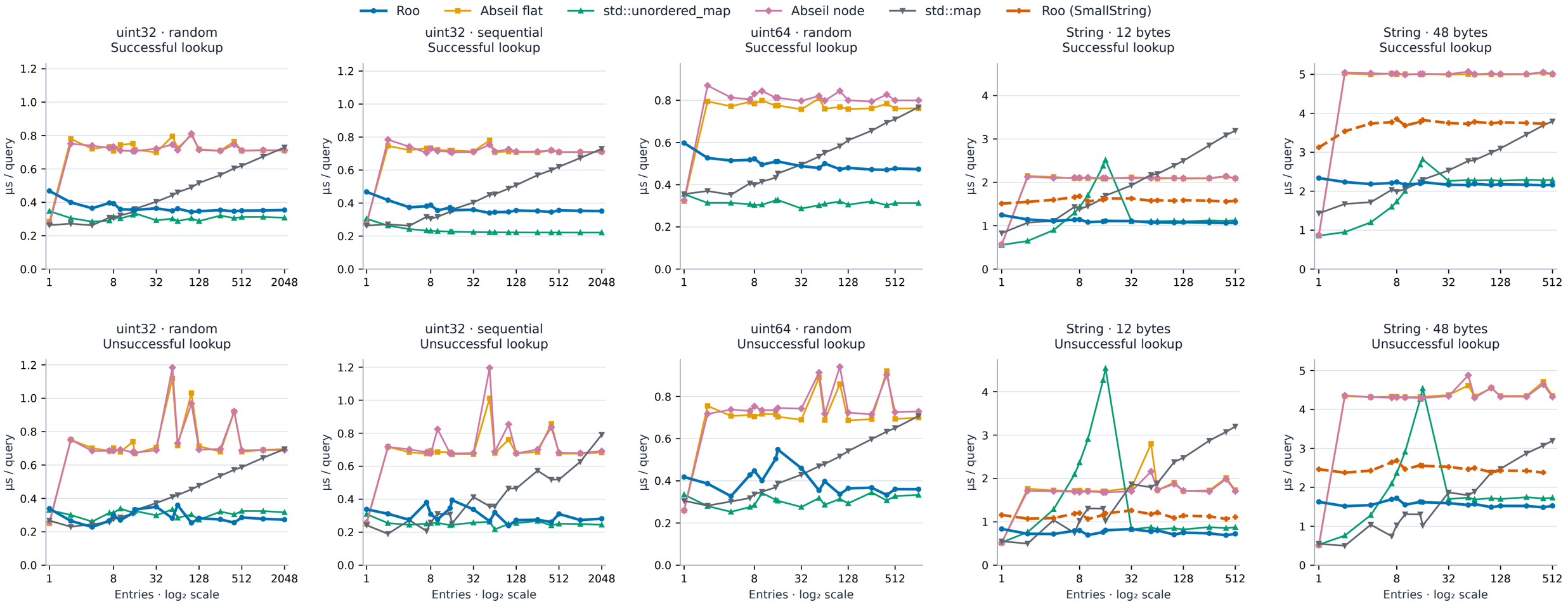
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Lookup performance across map sizes

Default hashers; std::map uses ordering. Y axes are linear and start at zero.



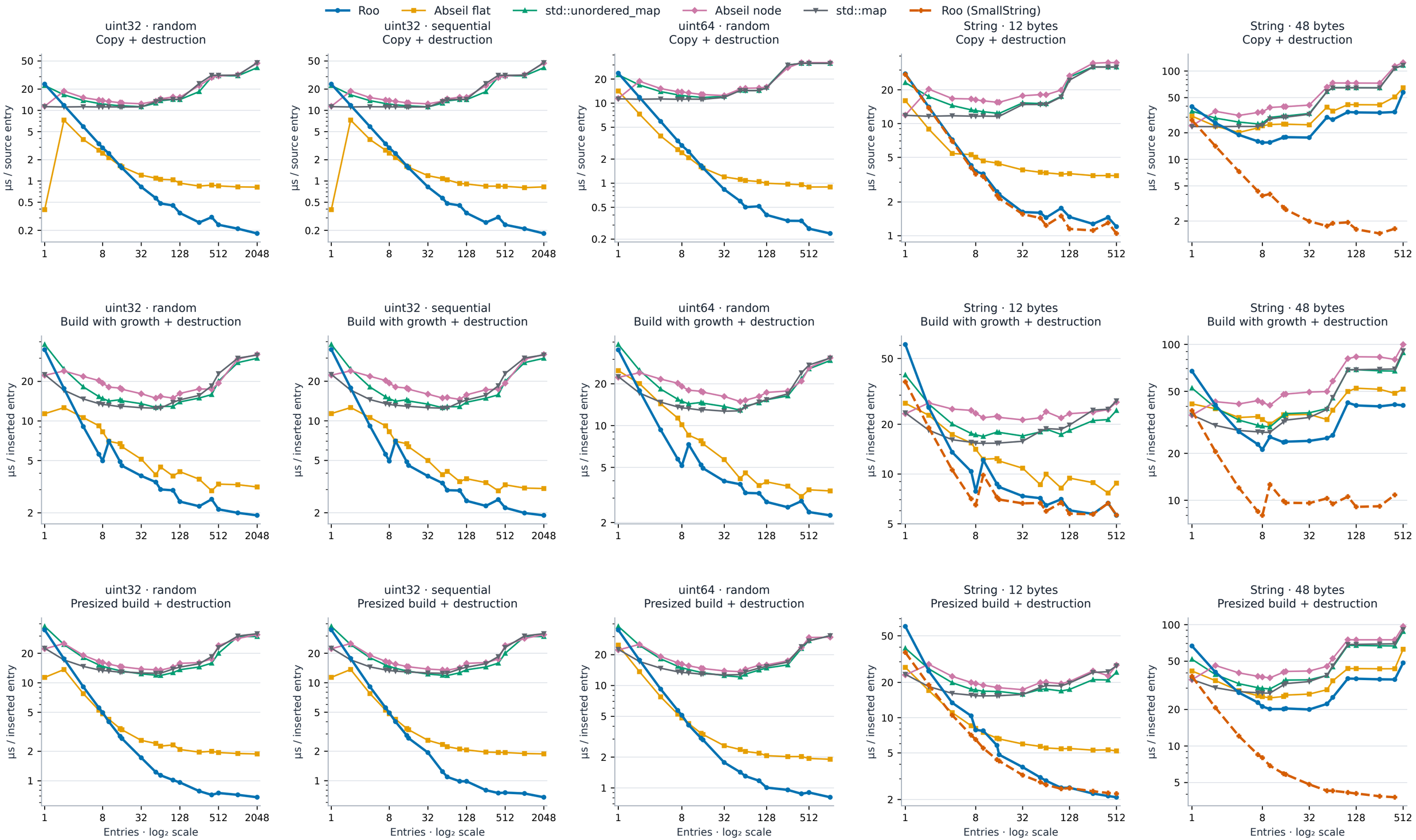
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Copy and construction costs

Default hashers; std::map uses ordering. Y axes are logarithmic.



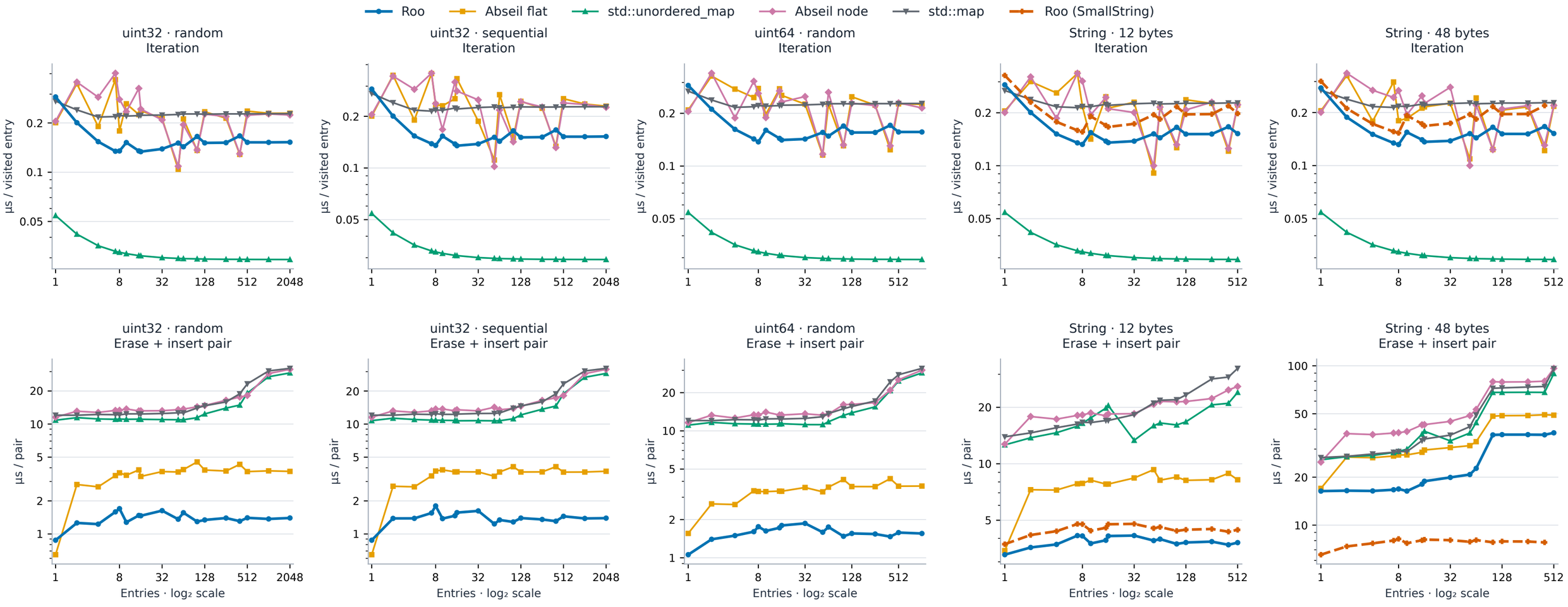
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Iteration and erase/insert costs

Default hashers; std::map uses ordering. Y axes are logarithmic.



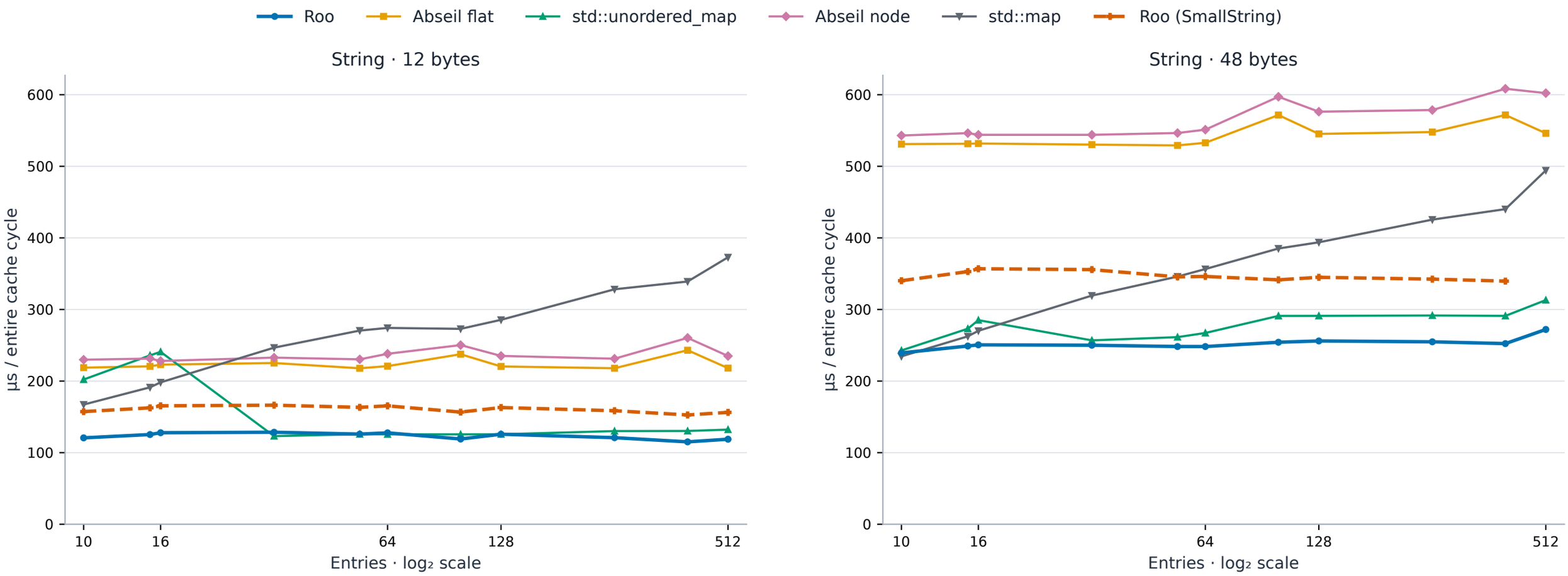
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Cache churn · string keys

Default hashers; std::map uses ordering. One cycle = 90 hits + 10 misses + one eviction + one insertion.



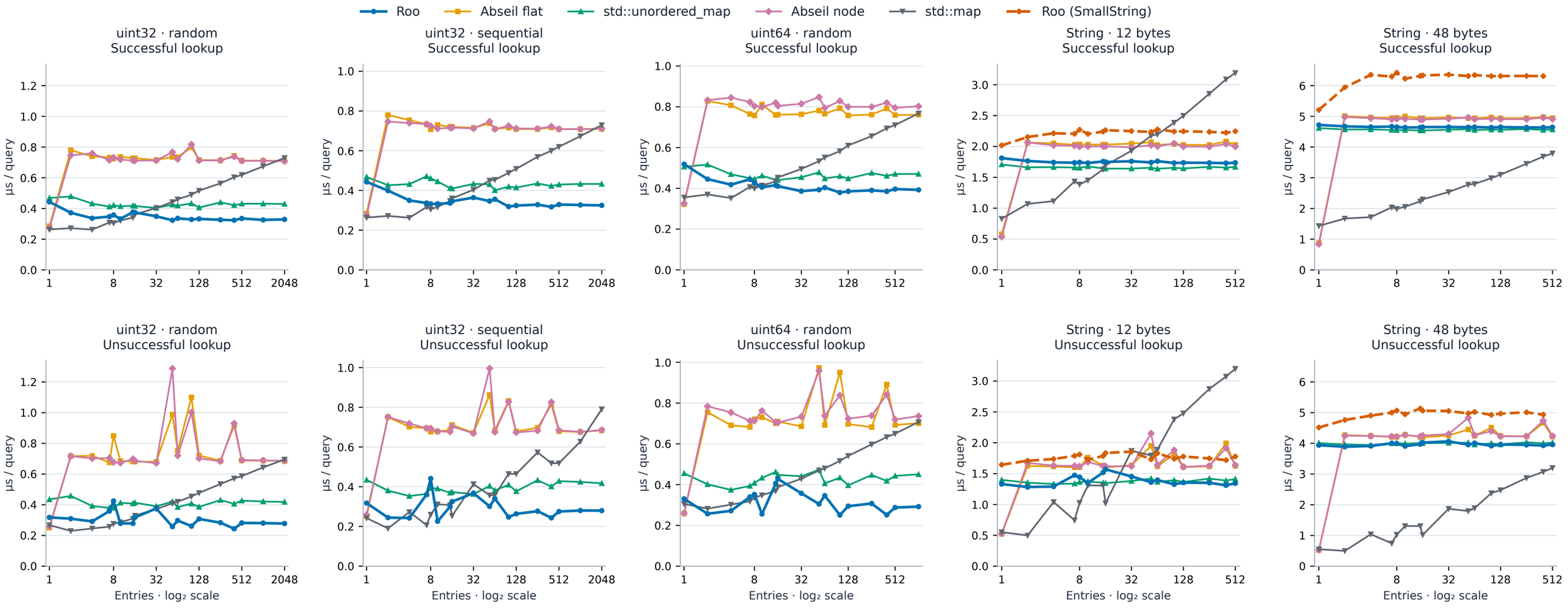
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Lookup performance across map sizes

Shared absl::Hash<Key> for every hash map; std::map uses ordering. Y axes are linear and start at zero.



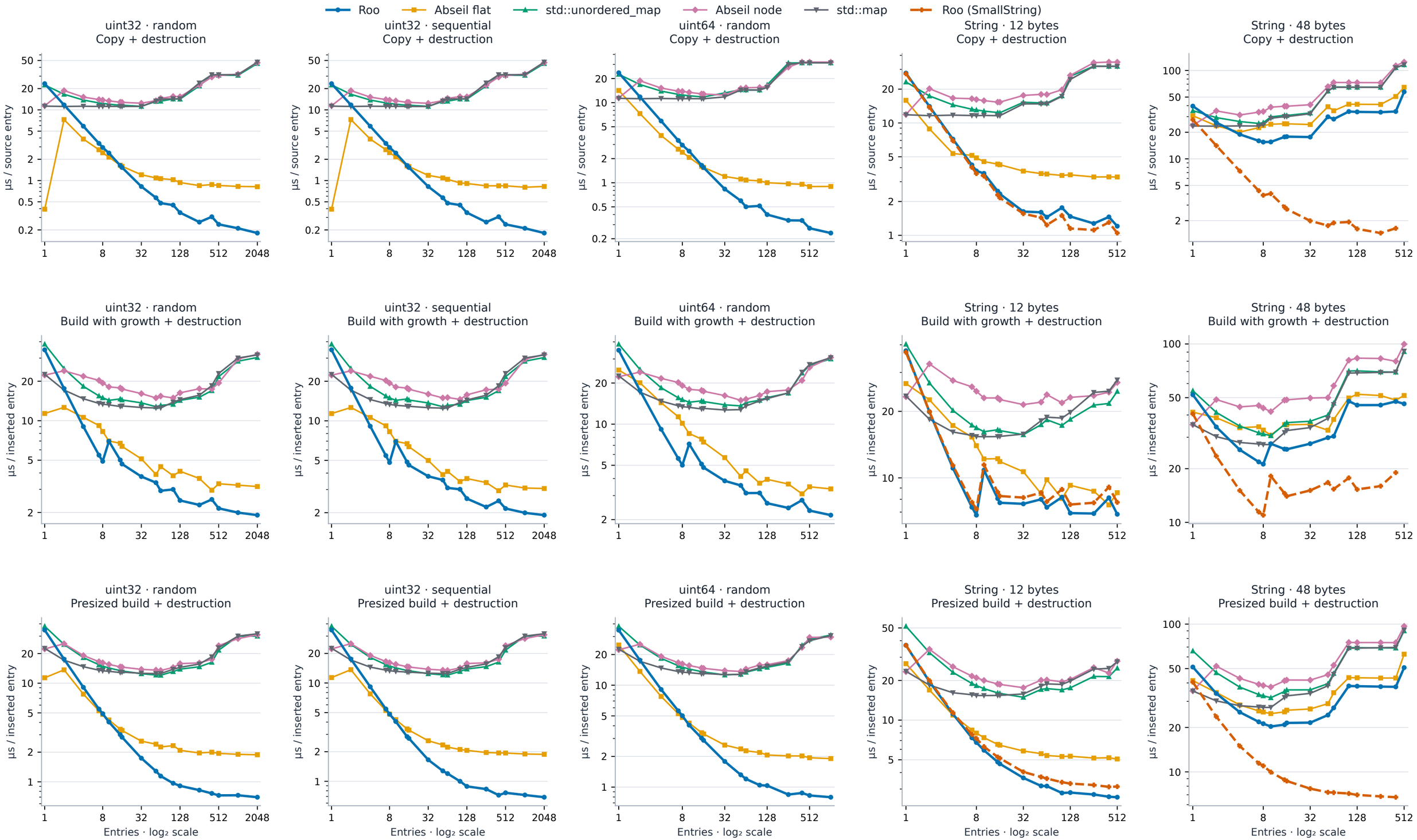
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Copy and construction costs

Shared absl::Hash<Key> for every hash map; std::map uses ordering. Y axes are logarithmic.



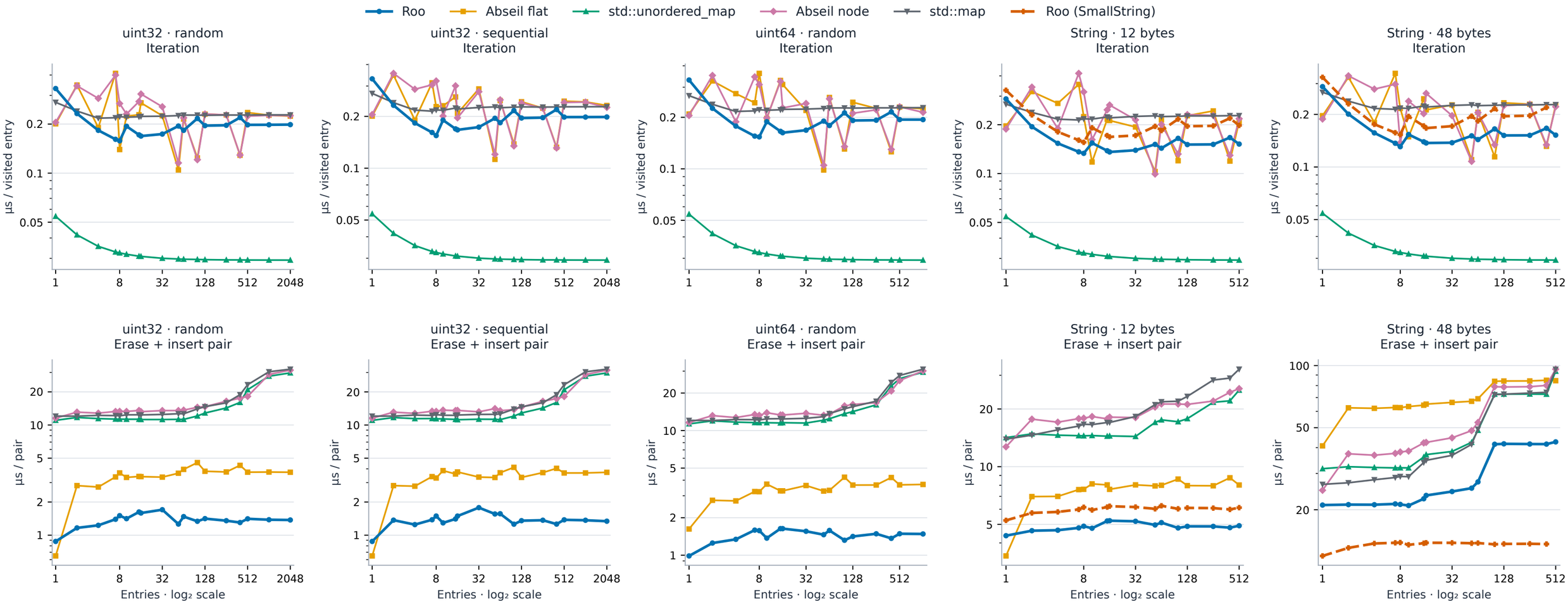
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Iteration and erase/insert costs

Shared absl::Hash<Key> for every hash map; std::map uses ordering. Y axes are logarithmic.



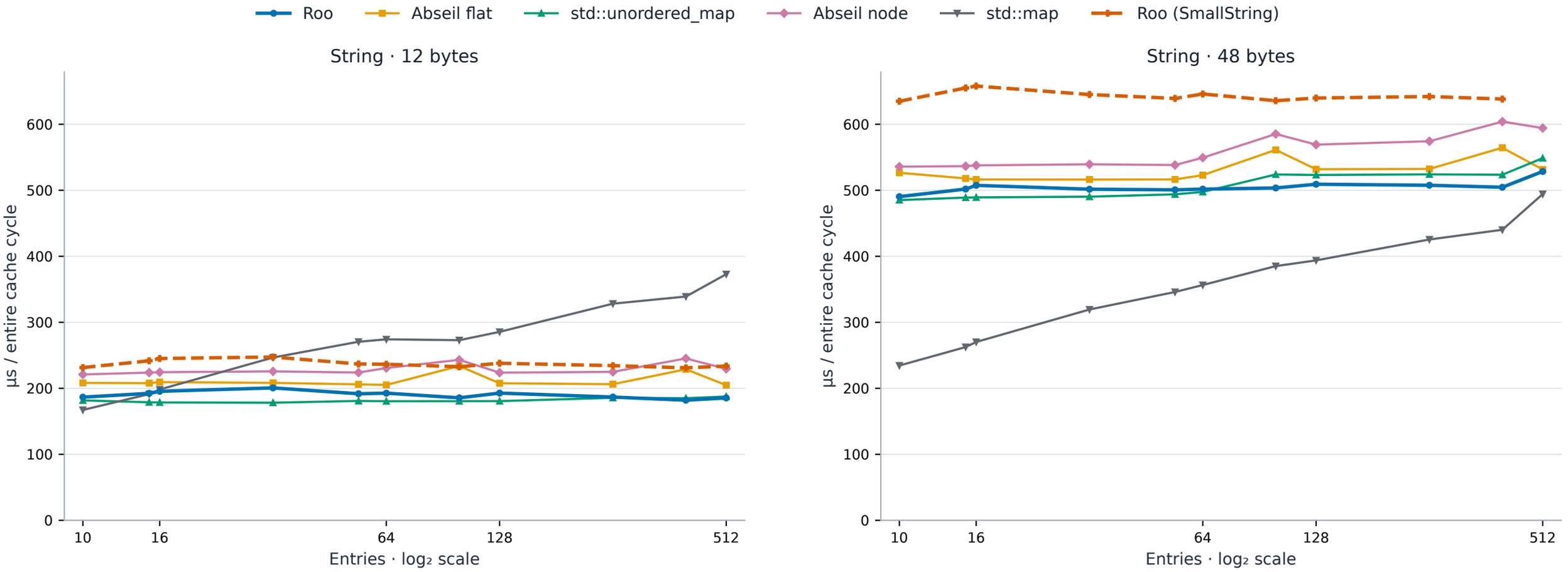
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Cache churn · string keys

Shared absl::Hash<Key>; std::map uses ordering. One cycle = 90 hits + 10 misses + one eviction + one insertion.



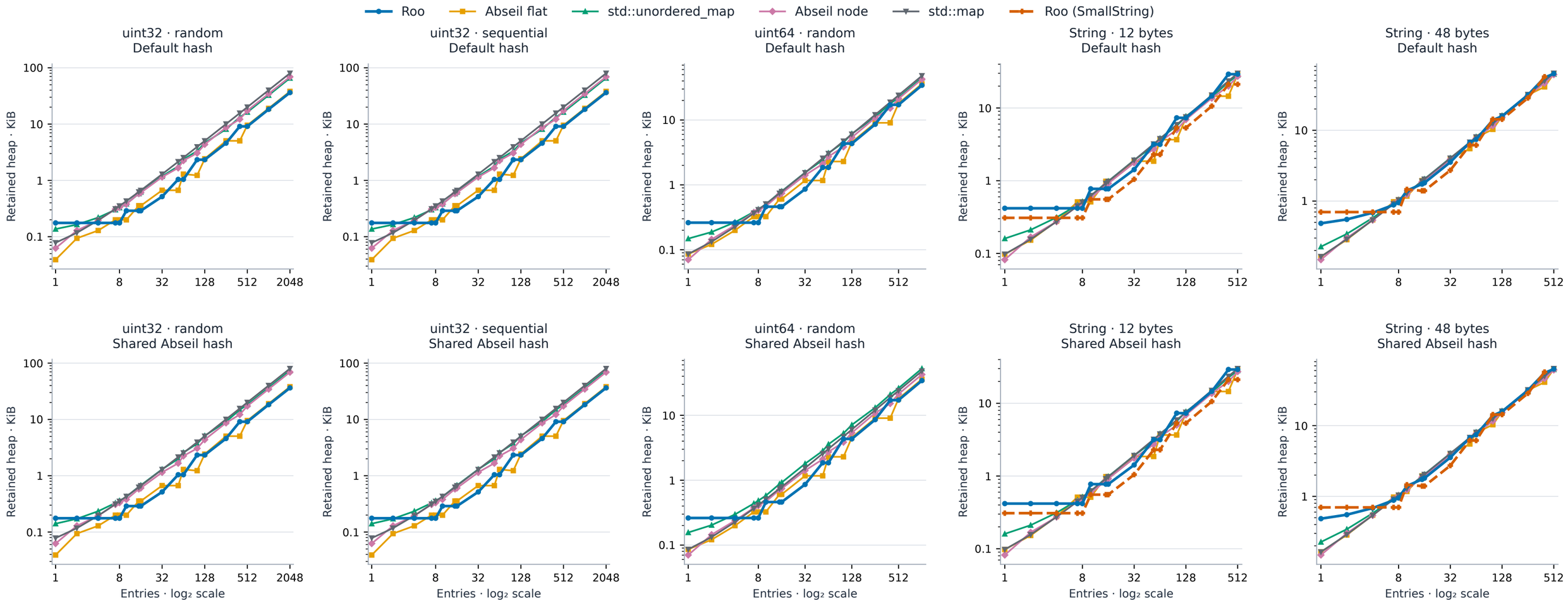
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Retained memory · default and shared hashing

Logarithmic axes. Grown-map footprint includes its object, owned strings and allocator overhead; not peak memory. std::map remains ordered.



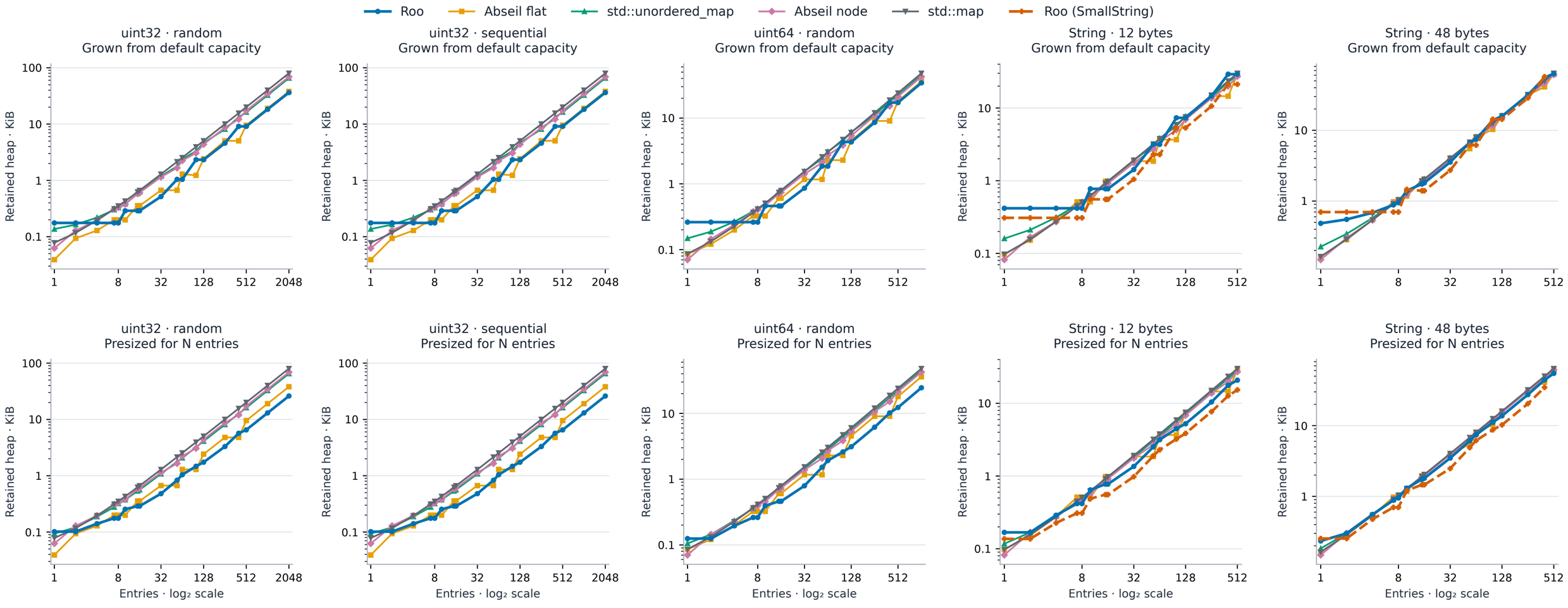
Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Retained memory · growing versus presizing

Default hashers; logarithmic axes. Heap includes the map object, owned strings and allocator overhead; not peak memory. std::map stays ordered.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: unrecorded · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.